

STAT3003: Survey Methods

2. Simple Random Sampling

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2.1 Probability vs Quota

- In the previous lecture, we said our sample would consist of elements **randomly chosen** from the population
- Samples based on planned randomness are called **probability samples**
- We saw that probability samples allowed us to establish estimator bias/unbiasedness and make **probability statements** about our estimators
 - E.g. confidence intervals

2.1 Probability vs Quota

- Why use probability samples? Why not just choose elements carefully to ensure a **representative** sample?
- Samples which select elements to ensure the sample has **very similar characteristics** as the population are called **quota samples**
 - E.g. select a sample which has the same proportions of sex, age, income, education etc. as the population
- Quota sampling was commonly used by **political pollsters** in the US up to and including the 1948 election

2.1 Probability vs Quota

Characteristic	Population Estimate (%)	Gallup Sample (%)	Roper Sample (%)
Sex – Male	49.1	50.5	50.1
Sex – Female	50.9	49.5	49.9
Age – 21-34	34.3	29.6	36.3
Age – 35-49	30.9	34.4	30.8
Age – 50+	34.8	35.9	32.9
Colour – White	96.1	95.0	97.4
Colour – Black	3.9	5.0	2.6
Etc.	⋮	⋮	⋮
# Respondents		2972	3501

Extract of Table 2.2 from Scheaffer et al. 2012: Comparison of Gallup and Roper samples with U.S. population

2.1 Probability vs Quota

- The 1948 presidential election was between Thomas Dewey and Harry Truman
- The Gallup and Roper polls used **quota sampling** to find samples similar to the overall population in terms of
 - Sex, age, education, colour, veteran status, labour union membership
- The polls predicted a comfortable victory for Dewey
 - Gallup had Dewey 49.5% to Truman's 44.5%
 - Roper had Dewey 52.2% to Truman's 37.1%

2.1 Probability vs Quota

- The result was Dewey 45.1% to Truman 49.5%
 - It seems the quota sampling dramatically overrepresented Dewey supporters (Republicans)
- How?
 - Instructions to interviewers in a quota sampling poll tended to be like “Find 2 men and 3 women in your block. Make sure 4 are over 25 years old and one is under 25.”
 - Instructions like these leave lots of room for human **subjectivity**. We want **objectivity**.
 - Perhaps the interviewer subconsciously (or consciously!) seeks out people who look “nicer” – e.g. richer and if Republicans tend to have higher incomes then...

2.1 Probability vs Quota

- Maybe the quota sampling wasn't specific enough
 - Why not include more characteristics, like income level, marital status, attitude towards issues like taxes, healthcare, foreign policy, etc.
 - Or combinations of the above, e.g. match the ratio of white, single males with union membership
- Clearly it is impossible to control all factors which may influence a person's preference of Dewey or Truman
 - Indeed, trying to fix one ratio might disturb other ratios
- Randomisation balances out all factors inherent in a population
 - including the factors you know but would like to control
 - those you know but can't control
 - and those you don't know you don't know

2.1 Probability vs Quota

- In this course, **all sample designs will be based on probability sampling**
 - All modern samples use probability sampling
- They produce more accurate results than quota sampling designs
- Also, they
 - Require a smaller sample size
 - Are easier to do
 - Are much more objective

2.1 Probability vs Quota

- What do you think of these sampling methods? (Exercise 4.5 in Scheaffer et al.)
 - A famous newspaper columnist estimates the proportion of unhappy marriages by asking her readers to write to her.
 - To find the average length of string in a bag, a student reaches in, mixes up the strings, selects one, mixes them up again, selects another, and so on

2.2 Simple Random Sampling

- This is our first **sample survey design**, i.e. a procedure for selecting a sample
- Let the population have size N
- **Simple Random Sampling** (SRS) is
 - Random sampling **without replacement** such that every possible sample of n units is **equally likely**
- Therefore, there are $\frac{N!}{n!(N-n)!}$ possible random samples and each occurs with the same probability

2.2 Simple Random Sampling

- Decide if the following sampling methods produce a SRS (Exercise 4.11 from Scheaffer et al.)
 - A class has 15 boys and 15 girls. Assign the boys numbers from 1 to 15, the girls from 16 to 30. Use a random number table to select six numbers from 1 to 30. Select the students with those numbers
 - Same class as above. This time use the random number table to select 3 numbers from 1 to 15 and 3 from 16 to 30
 - In a full class with six rows of five chairs each, randomly choose a row and select all students in that row

2.2 Simple Random Sampling

- Given a population of size N , what is the probability the i th element is in a SRS of size n ?
 - This is called the **inclusion probability** of element i
 - Let $\pi_i = P(\text{Element } i \text{ is in the sample})$
 - Then write **$\pi_i = E[Z_i]$** where Z_i is a random variable which equals 1 if element i is in the sample or 0 otherwise
 - $= \sum_{\substack{\text{All samples} \\ \text{containing element } i}} P(\text{Sample occurs})$
 - $= \sum_{\substack{\text{All samples} \\ \text{containing element } i}} \frac{1}{\frac{N!}{n!(N-n)!}}$

2.2 Simple Random Sampling

- Now say element i is in sample of size n
 - We have to fill the remaining $(n - 1)$ spaces from $(N - 1)$ elements in the population
 - There are $\left(\frac{(N-1)!}{(n-1)!(N-n)!}\right)$ ways to do this
 - Hence there are $\frac{(N-1)!}{(n-1)!(N-n)!}$ samples with element i
- Therefore, we have

$$- \pi_i = \frac{(N-1)!}{(n-1)!(N-n)!} \times \frac{1}{\frac{N!}{n!(N-n)!}} = \frac{n}{N} = E[Z_i]$$

2.2 Simple Random Sampling

- How to perform simple random sampling?
- Don't
 - Use your own judgement to “randomly” select (so-called **haphazard sampling**)
 - Try to select a “representative” sample
- Because
 - Both are prone to **investigator bias**
 - Both lead to estimators whose properties we don't know.

2.2 Simple Random Sampling

- How to perform simple random sampling?
- Do
 - Draw lots
 - Use random numbers in a book or Excel/R...
- Example
 - I want to draw a **SRS without replacement** of 10 students from a population of 21 students

2.2 Simple Random Sampling

- 1st Method: Draw Lots
 - Prepare 21 cards
 - Write a student number on each
 - Shuffle well
 - Pick up 10 cards

2.2 Simple Random Sampling

- 2nd Method: Use a [Random Numbers Table](#)
 - Find a page of random numbers
 - Assign the students numbers from 01 to 21
 - Randomly select a starting point in the random number table (drop a pencil, see where the pointed end points to)
 - Read off pairs of numbers in any direction you like
 - Sample without replacement

2.2 Simple Random Sampling

- 3rd Method: Use **Random Numbers in Excel/R...**
 - Can be done equivalently with other spreadsheet programs
 - Create a column of the 21 students ID numbers or names
 - Alongside, create a column of 21 **random numbers** using =RAND()
 - Use **Sort** to arrange the students names in order of ascending random numbers
 - Choose the first 10 names

2.2 Simple Random Sampling

- We've seen how to conduct simple random sampling
- What parameters might we be interested in?
We will consider three
 - Population mean μ
 - Population total τ
 - Population proportion p

2.2 Simple Random Sampling

- We shall see that τ and p can be written easily in terms of μ
- So we shall work out the following for μ and transfer the results to τ and p
 - What shall we use as an estimator $\hat{\mu}$ for μ ?
 - Is $\hat{\mu}$ unbiased?
 - What is the variance of $\hat{\mu}$?
 - How can we estimate the variance of $\hat{\mu}$?
 - How can we calculate confidence intervals for μ ?
 - How can we determine an appropriate sample size n for estimating μ ?

2.3 Properties of $\bar{Y}$

- Our estimator for μ is what we expect: **the sample mean**
- Let the population have values $\{u_1, u_2, \dots, u_N\}$
- Let $\{Y_1, Y_2, \dots, Y_n\}$ be the sample drawn by SRS without replacement
- The sample mean is given $\bar{Y} := \frac{1}{n} \sum_{i=1}^n Y_i$
 - We may write this another way: $\bar{Y} = \frac{1}{n} \sum_{j=1}^N u_j Z_j$

2.3 Properties of $\bar{Y}$

- We calculate $E(\bar{Y})$

$$\blacktriangleright = E\left(\frac{1}{n}\sum_{j=1}^N u_j Z_j\right) = \frac{1}{n}\sum_{j=1}^N u_j E(Z_j)$$

- Now, since $E(Z_j) = \frac{n}{N}$ for all j

$$\blacktriangleright E(\bar{Y}) = \frac{1}{n}\sum_{j=1}^N u_j \frac{n}{N} = \frac{1}{N}\sum_{j=1}^N u_j = \mu$$

- Thus $E(\bar{Y}) = \mu$ and our estimator is unbiased.
- This is good.

2.3 Properties of $\bar{Y}$

- Before we calculate $Var(\bar{Y})$, let us recall a useful result
 - Let $X_1, \dots, X_n$ be random variables and $a_1, \dots, a_n$ be constants
 - Then for $U := \sum_{i=1}^n a_i X_i$, we have
 - $Var(U) = \sum_{i=1}^n a_i^2 Var(X_i) + \sum_{i=1}^n \sum_{j \neq i} a_i a_j Cov(X_i, X_j)$
- Therefore, according to our useful result
 - $Var(\bar{Y}) = \sum_{j=1}^N \left(\frac{u_j}{n}\right)^2 Var(Z_j) + \sum_{j=1}^N \sum_{k \neq j} \frac{u_j}{n} \frac{u_k}{n} Cov(Z_j, Z_k)$
- Since Z_j is a Bernoulli random variable with probability of “success” $\frac{n}{N}$
 - $Var(Z_j) = \frac{n}{N} \left(1 - \frac{n}{N}\right)$

2.3 Properties of $\bar{Y}$

- Note
 - $Cov\left(Z_j, Z_k\right) = E\left(Z_j Z_k\right) - E\left(Z_j\right) E\left(Z_k\right) = E\left(Z_j Z_k\right) - \left(\frac{n}{N}\right)^2$
 - Note $Z_j Z_k$ equals 0 unless both elements j and k are in the sample, then $Z_j Z_k = 1$
 - The number of possible samples containing both j and k is $\frac{(N-2)!}{(n-2)!(N-n)!}$
 - Each SRS sample is equally likely to occur, with the number of possible random samples $\frac{N!}{n!(N-n)!}$
 - Hence $E\left(Z_j Z_k\right) = \frac{(N-2)!}{(n-2)!(N-n)!} \times \frac{1}{\frac{N!}{n!(N-n)!}} = \frac{n(n-1)}{N(N-1)}$

2.3 Properties of $\bar{Y}$

- Therefore

- $Cov(Z_j, Z_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)} = -\frac{n(1-\frac{n}{N})}{N(N-1)}$

- Thus

$$Var(\bar{Y}) = \frac{n}{N} \left(1 - \frac{n}{N}\right) \sum_{j=1}^N \frac{u_j^2}{n^2} - \frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{1}{N-1} \sum_{j=1}^N \sum_{k \neq j}^N \frac{u_j}{n} \frac{u_k}{n}$$
$$\Rightarrow Var(\bar{Y}) = \frac{1}{n^2} \frac{n}{N} \left(1 - \frac{n}{N}\right) \left[\sum_{j=1}^N u_j^2 - \frac{1}{N-1} \sum_{j=1}^N \sum_{k \neq j}^N u_j u_k \right]$$

2.3 Properties of $\bar{Y}$

- Now use the **identity**
 - $\sum_{j=1}^N (u_j - \mu)^2 = \frac{1}{N} [(N-1) \sum_{j=1}^N u_j^2 - \sum_{j=1}^N \sum_{j \neq k} u_j u_k]$
(exercise!)
 - $\Rightarrow \text{Var}(\bar{Y}) = \frac{1}{n^2} \frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{N}{N-1} \sum_{j=1}^N (u_j - \mu)^2$
 - $\Rightarrow \text{Var}(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$
- We have found the variance of our estimator
 - How is this different from the formulas you have seen before?

2.3 Properties of $\bar{Y}$

- We have found the variance of our estimator
- Notice the difference between this and the “usual” variance of the sample mean – there is a factor of $\left(\frac{N-n}{N-1}\right) = 1 - \frac{n-1}{N-1}$
- This is an example of a finite population correction, or a **f.p.c.**
 - We will also see the f.p.c. $\frac{N-n}{N} = 1 - \frac{n}{N}$ appear more commonly in our equations, which uses the **sampling fraction** $f = \frac{n}{N}$
- Thus a sample of 15 out of a population of 25 contains more information (less variability) than a sample of 15 out of a population of 25,000

2.3 Properties of $\bar{Y}$

- Unfortunately, most of the time **we do not know parameter σ**
- Can we estimate this as well?
- Let us use the **sample variance**
 - $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$
 - Denote by s^2 the **observed value** of $\hat{\sigma}^2$, i.e. $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$
- Is this an unbiased estimator for σ^2 ?

2.3 Properties of $\bar{Y}$

- By repeating a similar calculation to showing $Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$, we can show
 - $E[\hat{\sigma}^2] = \frac{N}{N-1} \sigma^2 = \frac{1}{1-\frac{1}{N}} \sigma^2$
 - (See Problem Sheet 1)
- Hence, $\hat{\sigma}^2$ is a biased estimator of σ^2
 - But $\left(1 - \frac{1}{N}\right) \hat{\sigma}^2$ is an unbiased estimator for σ^2

2.3 Properties of $\bar{Y}$

- Since $Var(\bar{Y}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)$, we see that
 - $\frac{1}{n} \frac{N-n}{N-1} \frac{N-1}{N} \hat{\sigma}^2 = \left(1 - \frac{n}{N} \right) \frac{1}{n} \hat{\sigma}^2$ is an unbiased estimator for $Var(\bar{Y})$
- As this quantity will be used rather often, we will give it its own notation:
 - $\widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N} \right) \frac{\hat{\sigma}^2}{n}$
 - Notice the appearance of the f.p.c.
- What happens as the population size grows large?

2.4 Confidence Intervals for μ

- Let's recap

- Our estimator $\hat{\mu}$ for parameter μ is the sample mean, $\bar{Y}$.
- We showed $\bar{Y}$ is unbiased. This means its Mean Square Error is equal to its variance.
- We found the variance of $\bar{Y}$. It depends on the population variance σ^2 .
- Since we usually do not know the true value of σ^2 , we will usually not know the true variance of $\bar{Y}$.
- So we found an unbiased estimator for the variance of $\hat{\mu}$.

2.4 Confidence Intervals for μ

- Knowing about the unbiasedness and variance of $\bar{Y}$ is indeed useful.
- But to create confidence intervals, we need to know about its **sampling distribution**
- Recall what we said before in “Introduction”. If the parameter is θ and the estimator is $\hat{\theta}$, it would be nice if
 - $\hat{\theta}$ is unbiased and
 - $\frac{(\hat{\theta} - E[\hat{\theta}])}{\sigma_{\hat{\theta}}} \sim N(0,1)$

2.4 Confidence Intervals for μ

- If $\hat{\theta}$ has these properties, then we can easily find a $100(1 - \alpha)\%$ confidence interval for θ
 - $\hat{\theta} \pm \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}$
- Let θ be μ and $\hat{\theta}$ be $\bar{Y}$
 - We have already shown $\bar{Y}$ is unbiased.
 - Can we say $\frac{\bar{Y} - \mu}{\sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}} \sim N(0,1)$?

2.4 Confidence Intervals for μ

- Yes, sometimes, according to the central limit theorem (CLT) for simple random sampling without replacement from finite populations Hájek (1960)
 - If certain technical conditions hold (!) and if n , N and $N - n$ are all sufficiently large,
 - then $\frac{\bar{Y} - \mu}{\sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}}$ is approximately $N(0,1)$ and
 - $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}$ is a large-sample $100(1 - \alpha)\%$ C.I.

2.4 Confidence Intervals for μ

- But usually we don't know the value of σ
 - Since $\frac{N-1}{N} \hat{\sigma}^2$ is an unbiased estimator for σ^2 , it is common to **replace σ** in the formula with $\sqrt{\frac{N-1}{N}} \hat{\sigma}$
 - And the C.I. becomes $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{\hat{\sigma}}{\sqrt{n}}$.
 - Note this is a random variable. The observed value is $\bar{y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{s}{\sqrt{n}}$.

2.4 Confidence Intervals for μ

- But hang on, that assumes
 - $\frac{\bar{Y}-\mu}{\sqrt{\frac{N-n}{N}\left(\frac{\hat{\sigma}^2}{n}\right)}} = \frac{\bar{Y}-\mu}{\sqrt{\widehat{Var}(\bar{Y})}} \sim N(0,1)$, which isn't true...
- However, we do know that if the $Y_1, Y_2, \dots, Y_n$ are i.i.d. normal, then
 - $\frac{\bar{Y}-\mu}{\sqrt{\widehat{Var}(\bar{Y})}} \sim t_{n-1}$ distribution
- Therefore, in practice we replace the $z_{\frac{\alpha}{2}}$ percentile with a $t_{n-1, \frac{\alpha}{2}}$ percentile, so the CI becomes
 - $\bar{y} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{s}{\sqrt{n}}$

2.4 Confidence Intervals for μ

- For large samples, $t_{n-1, 1-\frac{\alpha}{2}} \approx Z_{1-\frac{\alpha}{2}}$
- For smaller samples, $t_{n-1, 1-\frac{\alpha}{2}} > Z_{1-\frac{\alpha}{2}}$, so this t distribution CI will be larger
- Using these formulae for large populations correctly depends on two things
 - n
 - How “normal” the population is

2.4 Confidence Intervals for μ

- Some populations are quite normal, e.g. heights of statistics student
- Some populations are very skewed, e.g. income of statistics students 20 years after graduation
- So before you use the CLT (and those formulae for CIs), think about the normality of the population being studied
 - For the time being, it suffices to use the t distribution CIs. Later on, we will see what to do if we are seriously concerned about our normality assumption.

2.4 Confidence Intervals for μ

- We want to estimate the average amount of money students at a particular high school spend on junk food each day
 - There are 2010 students in the high school
 - We sample 100 by SRS
 - The total of observations in the sample is \$2907
 - The total of squared observations in the sample is 154593\$²
 - Find a 95% C.I. for μ
 - What happens if you treat the student population as infinite?

2.5 Sample sizes for μ

- We will say that our goal is to have a C.I. of a certain width, $2d$.
 - What sample size do I need to achieve this width?
- If $Var(\bar{Y})$ is known, the width of the $100(1 - \alpha)\%$ C.I. is
 - $2\sqrt{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} Z_{1-\frac{\alpha}{2}}$
- Hence the appropriate sample size is the n which solves
 - $\sqrt{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} Z_{1-\frac{\alpha}{2}} = d$
- Some calculation shows
 - $n = \frac{1}{\frac{1}{N} + \frac{d^2}{\sigma^2 Z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)}$

2.5 Sample sizes for μ

- The problem with this is... we usually don't know σ . How can we estimate it?
 - **Estimate it:** use the sample variance from your pilot survey (assuming you did one...)
 - **Copy it:** use some **historical data (existing studies)** that exists on the topic and plug-in values found there
 - **Guess it:** if you have a reasonable hypothesis for the distribution of the data, that can give you some information regarding the variance.
 - E.g. you believe the population is normal and you have an idea for the range of the data. Then you could estimate σ by $\frac{\text{Range}}{4}$ or $\frac{\text{Range}}{6}$ because 95% and 99.7% of values from a normal distribution are within 2 and 3 standard deviations of the mean respectively.

2.5 Sample sizes for μ

- Say pre-sample estimate for σ is v
- Then we may use this formula to find an appropriate sample size for us

- $$n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 z^2 \left(1 - \frac{\alpha}{2}\right)} \left(1 - \frac{1}{N}\right)}$$

2.5 Sample sizes for μ

- Hang on... isn't that cheating? What about the t -distribution?
 - Remember, we are finding a sensible value for n **before** we perform the survey
 - So just as we might not know σ , we cannot know what value s will take
 - The best we can do is plug-in an estimate from historical data, your pilot survey or one based on the range of values
- But you may feel that using the Normal distribution to find n is too aggressive
 - you would prefer to play safe and consider the t -distribution

2.5 Sample sizes for μ

- Then to find the appropriate sample size we need to solve

- $\sqrt{\left(1 - \frac{n}{N}\right) \frac{v^2}{n}} t_{n-1, 1-\frac{\alpha}{2}} = d$ for n

- i.e. to solve $n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$

- The difficulty here is that n appears on both sides of the equation
 - So **how can we solve this?**

2.5 Sample sizes for μ

- Use trial and error
- Or use Excel/R...
 - Let $N = 1000, v = 40, d = 10, \alpha = 0.05$
 - This table is taken from an Excel spreadsheet
 - We choose the first n which is larger than the right hand side of the equation
 - Thus, we choose $n = 61$

n	$t_{n-1, 1-\frac{\alpha}{2}}$	$\frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$
2	12.70620474	720.9169057
3	4.30265273	228.5171704
4	3.182446305	139.4499289
$\vdots$	$\vdots$	$\vdots$
60	2.000995378	60.20665759
61	2.000297822	60.16721346
62	1.999623585	60.12909785

2.6 Estimating τ

- The sweat and tears we have given working out results for estimating μ will now pay off
- Because τ is related to μ in such a simple way, we simply adapt all the formulae we developed for $\hat{\mu}$
- Note $\tau = \sum_{j=1}^N u_j = N \frac{1}{N} \sum_{j=1}^N u_j = N\mu$
- So we choose our estimator $\hat{\tau}$ to be $N\bar{Y}$

2.6 Estimating τ

- Is $\hat{\tau}$ unbiased?
 - $E(\hat{\tau}) = E(N\bar{Y}) = NE(\bar{Y}) = N\mu = \tau$. Good.
- What is the variance of $\hat{\tau}$?
 - $Var(\hat{\tau}) = Var(N\bar{Y}) = N^2 Var(\bar{Y}) = \frac{N-n}{N-1} \frac{N^2}{n} \sigma^2$
- How can we estimate the variance of $\hat{\tau}$?
 - Use $N^2 \widehat{Var}(\bar{Y}) = N^2 \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n} := \widehat{Var}(\hat{\tau})$

2.6 Estimating τ

- How can we calculate confidence intervals for τ ? **Just keep track of the N^2 ...**
 - If σ is known, we have the following $100 \times$
 $(1 - \alpha)\%$ C.I. for τ : $\left(\hat{\tau} \pm \sqrt{N^2 \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) Z_{1-\frac{\alpha}{2}}} \right)$
 - If σ is unknown, a $100 \times (1 - \alpha)\%$ C.I. for τ is
given by $\left(\hat{\tau} \pm \sqrt{N^2 \left(1 - \frac{n}{N} \right) \frac{s^2}{n} t_{n-1, 1-\frac{\alpha}{2}}} \right)$

2.6 Estimating τ

- How can we determine an appropriate sample size for estimating τ given τ 's C.I. of a certain width $2d$?
- Say, we have some number v as our pre-sample estimate for σ . We may solve

$$- n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 z^2 \left(1 - \frac{\alpha}{2}\right)}}$$

$$- \text{Or } n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 t^2 \left(1 - \frac{\alpha}{2}\right)}}$$

2.6 Estimating τ



- The Statistics Department is concerned that their students spend too much time watching the attractive football played by Tottenham Hotspur F.C.
- They would like to estimate the total time spent each week by students watching Tottenham
- The department has 750 students, from which a sample of 50 is taken. The average amount of time is 10.31 hours and $s^2 = 2.25$
- Estimate the total number of student-hours given to watching the Tottenham and give a 99% C.I.



2.7 Estimating p

- Recall p is the population proportion. This is just a special case of estimating the population mean.
- Each element in the population **either has the characteristic** (e.g. approves of the CEO's job performance; supports Tottenham Hotspur F.C., wears glasses, etc.) **or doesn't**
 - Let element j have value y_j , where $y_j = 1$ if element j has the characteristic, and $y_j = 0$ if not
- Then $p = \frac{1}{N} \sum_{j=1}^N y_j$. That is, **p is a special μ**

2.7 Estimating p

- Thus a good, unbiased estimator for p is
 - $\hat{p} = \frac{1}{n} \sum_{i=1}^n Y_i$, where $Y_i = 1$ if element i has the characteristic and $Y_i = 0$ otherwise
- To find $Var(\hat{p})$, use $Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$
 - Since each Y_i is a Bernoulli random variable with probability of success p , in this case $\sigma^2 = p(1 - p)$ (exercise!)
 - $\Rightarrow Var(\hat{p}) = \frac{N-n}{N-1} \frac{1}{n} p(1 - p)$
 - Note this contains the parameter we want to estimate...

2.7 Estimating p

- To estimate $Var(\hat{p})$,
 - consider $\widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}$
 - when the Y_i are Bernoulli random variables, $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 = \frac{n}{n-1} \hat{p} (1 - \hat{p})$ (exercise!)
 - $\Rightarrow \widehat{Var}(\hat{p}) = \left(1 - \frac{n}{N}\right) \frac{1}{n-1} \hat{p}(1 - \hat{p})$ is an unbiased estimator for $Var(\hat{p})$

2.7 Estimating p

- To calculate a $100(1 - \alpha)\%$ confidence interval for p , assuming n is large enough, we must use
 - $\left(\hat{p} \pm \sqrt{\left(1 - \frac{n}{N}\right) \frac{1}{n-1} \hat{p}(1 - \hat{p})} t_{n-1, 1-\frac{\alpha}{2}} \right)$
 - Because $\sigma^2 = p(1 - p)$ will be unknown. If it is known we would know p
- What is large enough, in this case?
 - We require $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$

2.7 Estimating p

- Adapting the formulae for finding sample size n , we have

- $$n = \frac{1}{\frac{1}{N} + \frac{d^2}{p(1-p)z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)}$$

- Obviously, this formula contains p , which we don't know. What can we plug in?
 - An “educated guess” from historical data
 - For a conservative sample size, use $p = 0.5$. ($p(1 - p)$ is maximized at $p = 0.5$)

2.7 Estimating p

- We want to estimate the proportion of statistics student affected a disease called gooneritis
 - There are 1500 students
 - We know that about 30% of maths students suffer from gooneritis
 - We plan to estimate p using SRS.
 - What sample size do I need so that the total length of a 95% C.I. is less than 0.02?
 - What should I do if I think statistics students are very different to maths students?

2.8 Using SRS in Practice

- SRS is our first and simplest survey design. It may not be the best choice for your project
 - Are you sure you want a survey? Perhaps an **experiment** would be better?
 - What if you **do not have a frame**?
 - What if SRS is **too expensive/time-consuming** to perform?
 - What if you know the population you want to study has **certain characteristics** that could help you design a more effective sampling scheme?

2.8 Using SRS in Practice

- On the other hand, SRS is a good choice if
 - You have **very little information** about the population other than a frame
 - The person paying you to do the survey **insists you use SRS**. For example, in court cases SRS is often the method of choice because it is “simple” enough for jurors to understand